Check Point 6

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1 Introductions

This analysis uses data from the National Health and Nutrition Examination Survey (NHANES) Vision and Eye Health Surveillance dataset. NHANES is a nationally representative survey of the U.S. civilian, non-institutionalized population. The dataset includes demographic variables (e.g., Sex) and vision-related prevalence estimates (Data_Value) across survey cycles.

Research Question: Are prevalence estimates of eye health conditions associated with age and race/ethnicity among U.S. adults?

2 Material & Methods

2.1 Data

```
# Packages
library(tidyverse)
library(knitr)

# Set working directory
setwd("/Users/lynnliu/Desktop/R Data/CP6")

df <- read_csv(
  "National_Health_and_Nutrition_Examination_Survey_(NHANES)_-_Vision_and_Eye_Health_Surveillance_20260
)
```

```r
# Remove suppressed values and missing Data_Value
df <- as.data.frame(df)
df <- df[is.na(df$Data_Value_Footnote_Symbol), ]
df <- df[!is.na(df$Data_Value), ]

# Subset: exam-based conditions only, national level,
# all participants (no specific risk factor), both sexes,
# exclude "All races" and "All ages" so we get group-level estimates
eye_data <- df[
  df$Topic         == "Eye Health Conditions" &
  df$RiskFactor    == "All participants" &
  df$Sex           == "Both sexes" &
  df$RaceEthnicity != "All races" &
  df$Age           != "All ages",
]

# Keep relevant columns
eye_data <- eye_data[, c("Category", "Age", "RaceEthnicity",
                         "Data_Value", "Low_Confidence_limit",
                         "High_Confidence_Limit")]

eye_data$Age <- factor(eye_data$Age,
                       levels = c("40-64 years", "65-79 years",
                                  "80 years and older"))

eye_data$RaceEthnicity <- factor(eye_data$RaceEthnicity,
                                 levels = c("White, non-Hispanic",
                                            "Black, non-Hispanic",
                                            "Hispanic, any race",
                                            "Other"))
nrow(eye_data)
head(eye_data)
```

```r
# Descriptive stats by Age
eye_data <- eye_data[!is.na(eye_data$Age), ]
eye_data <- eye_data[eye_data$RaceEthnicity != "Other", ]
eye_data$RaceEthnicity <- droplevels(eye_data$RaceEthnicity)
age_desc <- do.call(rbind, lapply(
  levels(eye_data$Age), function(a) {
    x <- eye_data$Data_Value[eye_data$Age == a]
    data.frame(Age = a, Mean = round(mean(x),1), SD = round(sd(x),1),
               Min = round(min(x),1), Max = round(max(x),1))
  }
))
kable(age_desc, caption = "Prevalence (%) of eye conditions by age group")
```

Table 1: Prevalence (%) of eye conditions by age group

Age	Mean	SD	Min	Max
40-64 years	7.8	8.3	0.9	26.9
65-79 years	13.2	8.0	2.6	27.2

Age	Mean	SD	Min	Max
80 years and older	19.4	8.1	8.5	36.1

```r
# Descriptive stats by Race/Ethnicity
race_desc <- do.call(rbind, lapply(
  levels(eye_data$RaceEthnicity), function(r) {
    x <- eye_data$Data_Value[eye_data$RaceEthnicity == r]
    data.frame(RaceEthnicity = r, Mean = round(mean(x),1), SD = round(sd(x),1),
               Min = round(min(x),1), Max = round(max(x),1))
  }
))
kable(race_desc, caption = "Prevalence (%) of eye conditions by race/ethnicity")
```

Table 2: Prevalence (%) of eye conditions by race/ethnicity

RaceEthnicity	Mean	SD	Min	Max
White, non-Hispanic	13.1	9.5	1.3	36.1
Black, non-Hispanic	12.7	9.6	1.3	26.9
Hispanic, any race	10.4	8.0	0.9	25.1

2.2 Statistical Analysis

```r
#Normality CHeck - Age
shapiro_age <- do.call(rbind, lapply(
  levels(eye_data$Age), function(a) {
    x <- eye_data$Data_Value[eye_data$Age == a]
    if (length(x) >= 3) {
      s <- shapiro.test(x)
      data.frame(Group = a, W = round(s$statistic, 3), p.value = round(s$p.value, 4))
    } else {
      data.frame(Group = a, W = NA, p.value = NA)
    }
  }
))
kable(shapiro_age, caption = "Shapiro-Wilk test by age group")
```

Table 3: Shapiro-Wilk test by age group

	Group	W	p.value
W	40-64 years	0.790	0.0020
W1	65-79 years	0.923	0.1651
W2	80 years and older	0.916	0.3954

```r
#Normaility Check - Race/Ethnicity
shapiro_race <- do.call(rbind, lapply(
  levels(eye_data$RaceEthnicity), function(r) {
    x <- eye_data$Data_Value[eye_data$RaceEthnicity == r]
```

```r
    if (length(x) >= 3) {
      s <- shapiro.test(x)
      data.frame(Group = r, W = round(s$statistic, 3), p.value = round(s$p.value, 4))
    } else {
      data.frame(Group = r, W = NA, p.value = NA)
    }
  }
))
kable(shapiro_race, caption = "Shapiro-Wilk test by race/ethnicity")
```

Table 4: Shapiro-Wilk test by race/ethnicity

	Group	W	p.value
W	White, non-Hispanic	0.934	0.2524
W1	Black, non-Hispanic	0.891	0.0830
W2	Hispanic, any race	0.905	0.2487

```r
#Two-way ANOVA (Age + Race/Ethnicity)
anova_both <- aov(Data_Value ~ Age + RaceEthnicity, data = eye_data)
anova_summary <- as.data.frame(summary(anova_both)[[1]])
anova_summary <- round(anova_summary, 4)
kable(anova_summary, caption = "Two-way ANOVA: Age and Race/Ethnicity")
```

Table 5: Two-way ANOVA: Age and Race/Ethnicity

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Age	2	743.2281	371.6141	5.3329	0.0094
RaceEthnicity	2	21.1907	10.5954	0.1520	0.8595
Residuals	36	2508.6016	69.6834	NA	NA

```r
#Post-hoc Bonferroni - Age
age_posthoc <- pairwise.t.test(eye_data$Data_Value,
                               eye_data$Age,
                               p.adjust.method = "bonferroni")
kable(age_posthoc$p.value, caption = "Post-hoc Bonferroni: Age")
```

Table 6: Post-hoc Bonferroni: Age

	40-64 years	65-79 years
65-79 years	0.1982594	NA
80 years and older	0.0064949	0.2463593

```r
# Post-hoc Bonferroni - Race/Ethnicity
race_posthoc <- pairwise.t.test(eye_data$Data_Value,
                                eye_data$RaceEthnicity,
                                p.adjust.method = "bonferroni")
kable(race_posthoc$p.value, caption = "Post-hoc Bonferroni: Race/Ethnicity")
```

Table 7: Post-hoc Bonferroni: Race/Ethnicity

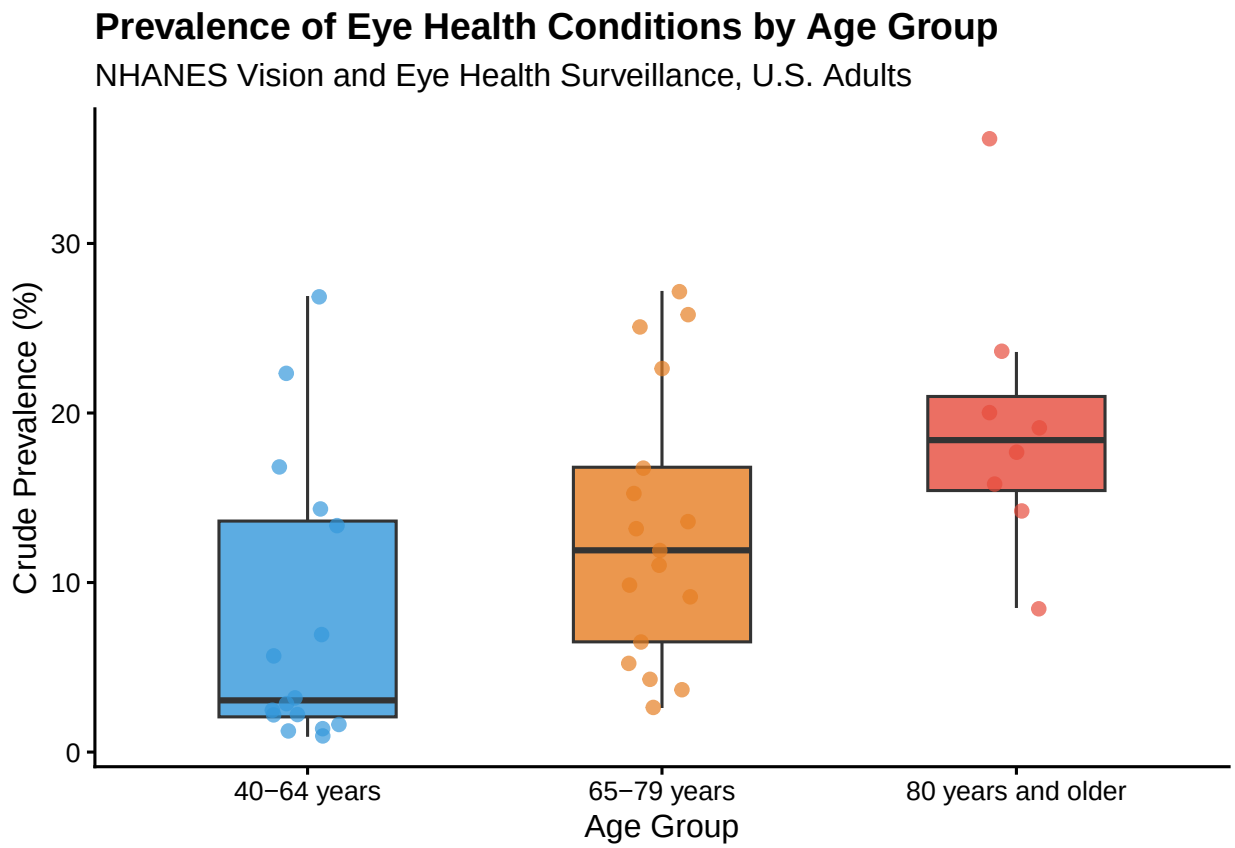
	White, non-Hispanic	Black, non-Hispanic
Black, non-Hispanic	1	NA
Hispanic, any race	1	1

2.3 Code

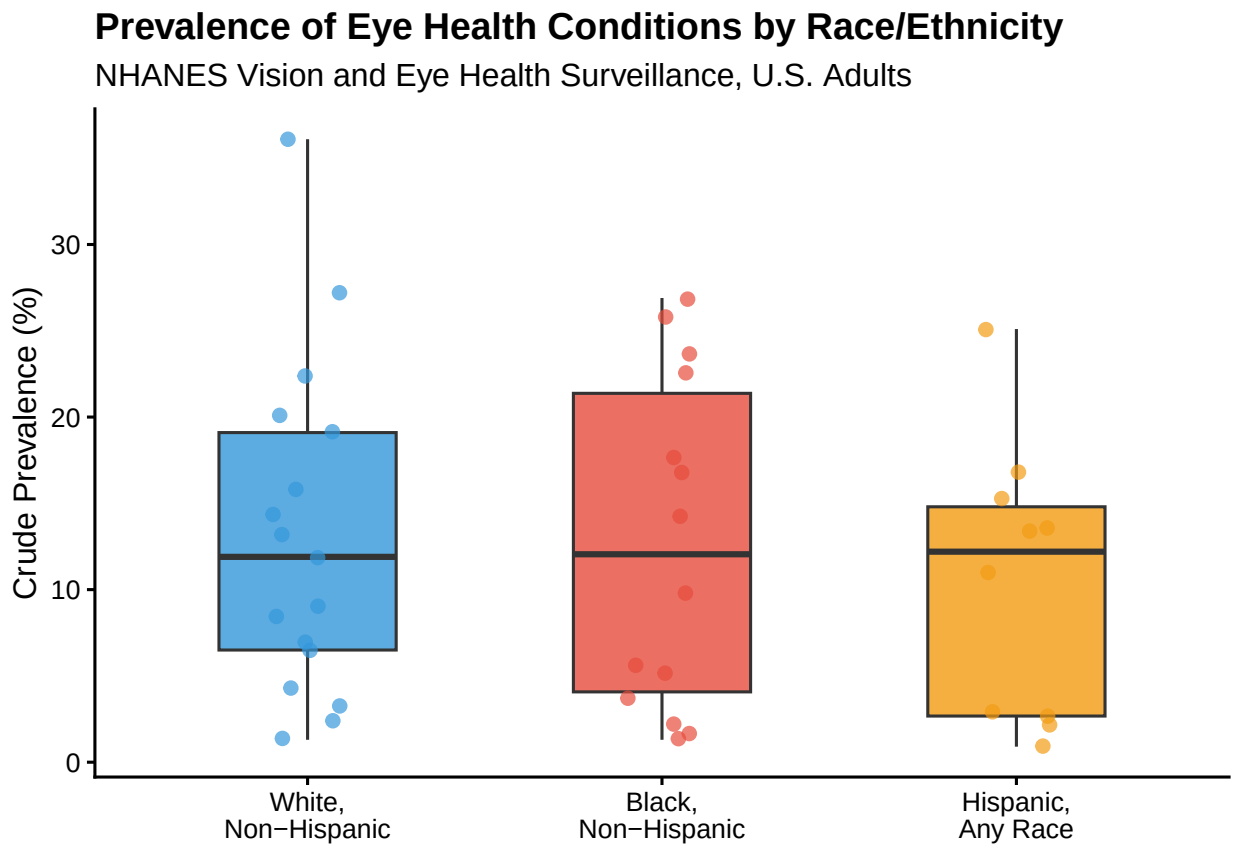
All code and data used in this analysis are available at: https://github.com/wl826-boop/R-Class

3 Results

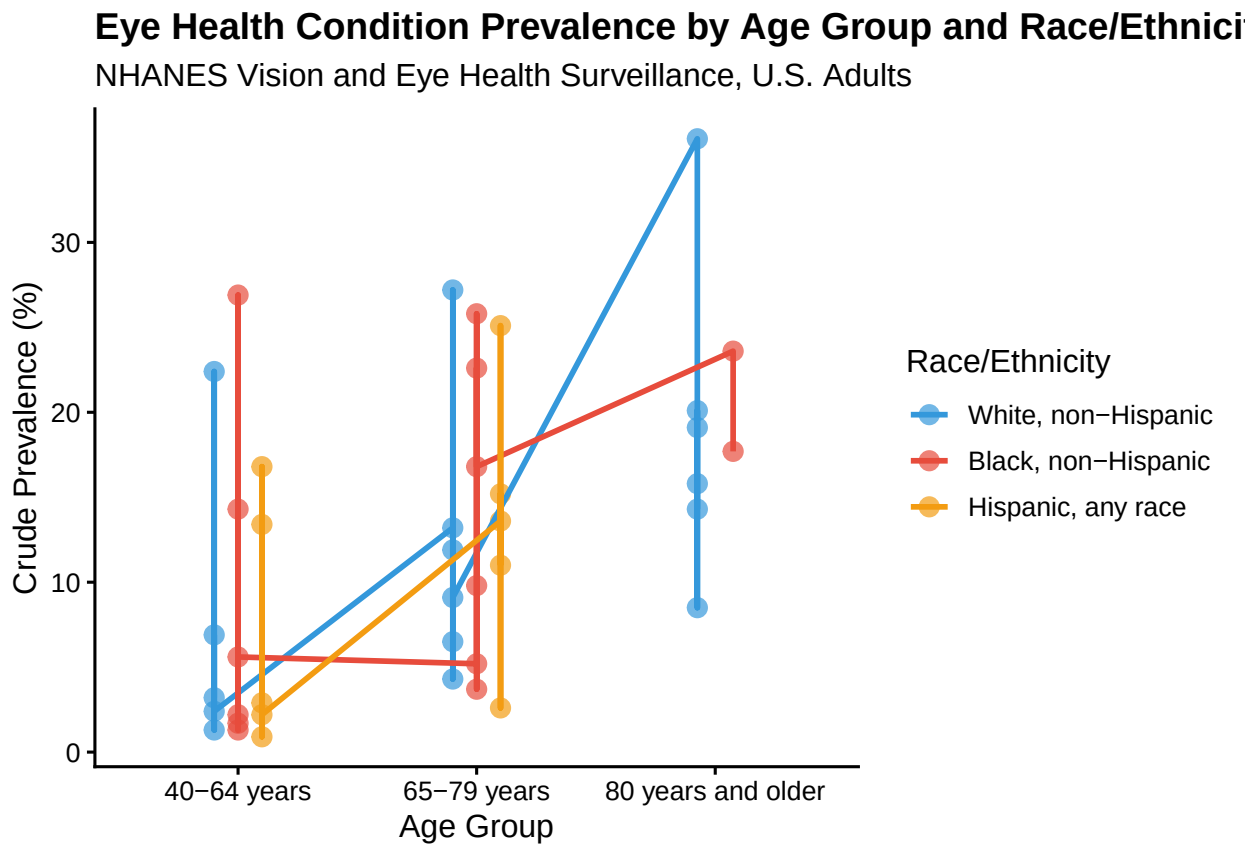
```
pal_age  <- c("40-64 years"        = "#3498DB",
              "65-79 years"        = "#E67E22",
              "80 years and older" = "#E74C3C")

pal_race <- c("White, non-Hispanic" = "#3498DB",
              "Black, non-Hispanic" = "#E74C3C",
              "Hispanic, any race"  = "#F39C12",
              "Other"               = "#2ECC71")

#Figure 1: Prevalence by Age Group
ggplot(eye_data, aes(x = Age, y = Data_Value, fill = Age)) +
  geom_boxplot(alpha = 0.8, outlier.shape = NA, width = 0.5) +
  geom_jitter(aes(color = Age), width = 0.1, size = 2, alpha = 0.7) +
  scale_fill_manual(values = pal_age) +
  scale_color_manual(values = pal_age) +
  labs(
    title    = "Prevalence of Eye Health Conditions by Age Group",
    subtitle = "NHANES Vision and Eye Health Surveillance, U.S. Adults",
    x        = "Age Group",
    y        = "Crude Prevalence (%)"
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold"))
```




```r
#Figure 2: Prevalence by Race/Ethnicity
ggplot(eye_data, aes(x = RaceEthnicity, y = Data_Value, fill = RaceEthnicity)) +
  geom_boxplot(alpha = 0.8, outlier.shape = NA, width = 0.5) +
  geom_jitter(aes(color = RaceEthnicity), width = 0.1, size = 2, alpha = 0.7) +
  scale_fill_manual(values = pal_race) +
  scale_color_manual(values = pal_race) +
  scale_x_discrete(labels = c("White,\nNon-Hispanic", "Black,\nNon-Hispanic",
                              "Hispanic,\nAny Race", "Other")) +
  labs(
    title    = "Prevalence of Eye Health Conditions by Race/Ethnicity",
    subtitle = "NHANES Vision and Eye Health Surveillance, U.S. Adults",
    x        = NULL,
    y        = "Crude Prevalence (%)"
  ) +
  theme_classic(base_size = 12) +
  theme(legend.position = "none",
        plot.title = element_text(face = "bold"))
```




```r
#Figure 3: Age x Race/Ethnicity interaction
ggplot(eye_data, aes(x = Age, y = Data_Value,
                     color = RaceEthnicity, group = RaceEthnicity)) +
  geom_point(size = 3, alpha = 0.7,
             position = position_dodge(width = 0.3)) +
  geom_line(linewidth = 1,
            position = position_dodge(width = 0.3)) +
  scale_color_manual(values = pal_race) +
  labs(
    title    = "Eye Health Condition Prevalence by Age Group and Race/Ethnicity",
    subtitle = "NHANES Vision and Eye Health Surveillance, U.S. Adults",
    x        = "Age Group",
    y        = "Crude Prevalence (%)",
    color    = "Race/Ethnicity"
  ) +
  theme_classic(base_size = 12) +
  theme(plot.title      = element_text(face = "bold"),
        legend.position = "right")
```



4 Conclusion

The significant association between age and eye condition prevalence (F = 5.33, p = 0.009) makes sense — older adults have had more years of exposure to risk factors and natural physiological decline, which explains why prevalence nearly tripled from the 40–64 group (7.8%) to the 80+ group (19.4%). The null finding for race/ethnicity (p = 0.859) was surprising given the research question, but the similar means across White (13.1%), Black (12.7%), and Hispanic (10.4%) groups suggest that when you pool all eye conditions together the racial differences may wash out. A more targeted analysis looking at specific conditions like glaucoma or diabetic retinopathy separately would likely tell a different story.

5 AI Use Disclosure Statement

This document was generated using ChatGPT to assist with understanding the assignment instructions and drafting initial R code for data subsetting, visualization, and probability calculations.